INSTITUTE OF TECHNOLOGY, NIRMA UNIVERSITY

Computer Vision · Edge AI · Smart Traffic Systems

This poster presents a **systematic comparison** of camera-based sensor technologies used in traffic monitoring systems. The objective is to identify the most suitable camera system for detecting violations such as **helmetless riding** using IoT and computer vision techniques.

CAMERA SENSOR OVERVIEW

Raspberry Pi Camera / Webcam

MEDIUM

PRINCIPLE
Images/video processed via AI on Raspberry Pi

PROS
Low cost, easy setup, flexible for prototyping

CONS
Low FPS, limited processing, affected by lighting

CCTV Camera

HIGH

PRINCIPLE
Continuous video capture processed on external server

PROS
Reliable, widely used, good coverage

CONS
Requires backend processing, no built-in intelligence

IP Camera (Smart Camera)

HIGH

PRINCIPLE
Network-enabled with streaming and remote access

PROS
High resolution, scalable, remote monitoring

CONS
Higher cost than basic cameras

AI Edge Camera (Jetson / Smart AI)

VERY HIGH

PRINCIPLE
Onboard AI processing for real-time detection

PROS
Real-time, high accuracy, reduces latency

CONS
Expensive, higher power consumption

Thermal Camera

LOW

PRINCIPLE
Detects heat signatures instead of visible light

PROS
Works in low light / night conditions

CONS
Cannot clearly detect helmets or visual details

SENSOR TYPE	WORKING PRINCIPLE	ADVANTAGES	LIMITATIONS	SUITABILITY
Raspberry Pi Camera / Webcam	Images/video processed using AI on Raspberry Pi	Low cost, easy to set up, flexible for prototyping	Low FPS, limited processing power, affected by lighting	MEDIUM
CCTV Camera	Continuous video capture, processed on external server	Reliable, widely used, good coverage	Requires backend processing, no built-in intelligence	HIGH
IP Camera (Smart Camera)	Network-enabled camera with streaming and remote access	High resolution, scalable, remote monitoring	Higher cost than basic cameras	HIGH
AI Edge Camera (Jetson)	Camera with onboard AI processing for real-time detection	Real-time processing, high accuracy, reduces latency	Expensive, higher power consumption	VERY HIGH
Thermal Camera	Detects heat signatures instead of visible light	Works in low light/night conditions	Cannot clearly detect helmets or visual details	LOW

⚠ LIMITATIONS

- Raspberry Pi camera has low FPS and limited processing capability
- Performance drops significantly in low light conditions
- Not ideal for large-scale deployment scenarios

◈ JUSTIFICATION

Camera-based systems are essential because helmet detection is a *visual classification problem* requiring image/video data. Other sensors like ultrasonic, IR, RFID, GPS, and LiDAR cannot replace cameras — they do not provide visual information.

✓ CONCLUSION

AI-enabled camera systems, especially **IP cameras and edge AI cameras**, provide the most effective and scalable solution for real-time traffic violation detection in IoT environments.